

Shared Building Blocks

The common parts every screen leans on

⚠ FOR UNDERSTANDING PURPOSES ONLY — NOT FOR COMMERCIAL OR TEACHING USE

Prasanaa Kartick T — Personal App Project

Flutter / Dart · Android

 Storage

 Sound

 Widgets

 Pickers

1. Why Shared Parts Matter

DepoVaultTick has dozens of screens, but they all feel like one app because they share the same few building blocks. Fix or improve one of these and the whole app benefits.

2. The Filing Cabinet (Storage)

One central service knows how to save every list of records to the phone and load it back. **Screens never talk to storage directly.** Every Add, Edit or Delete eventually calls it, and every screen calls its matching “load” when it opens.

- It covers family members (with their banks and deposits), Stash cards, notes, mutual funds, insurance policies, Post Office entries, saved invoices, work contacts, certificates and folders.
- Records are converted to simple text for storage and rebuilt when loaded. Anything missing or damaged falls back to a safe default instead of crashing.
- Nothing saves automatically in the background — each save happens because a screen explicitly asked, right after the user made a change.

3. The Data Models in One Table

Record	In plain terms
Family member	A person; holds banks; computes totals across them
Bank	A named folder of one person’s deposits; computes totals and “next closing”
Deposit	One FD / RD receipt
Insurance policy	One policy, with top-up payments
Post Office entry	One passbook page, with top-ups and a due-day pattern

Record	In plain terms
Mutual fund entry	A SIP (with pause history) or a lump sum
Note	A sticky note with styled text runs and a colour
Stash item	A titled secret
Work contact	A Rolodex card
Certificate · Certificate folder	A licence with expiry and reminder; a nestable folder
Saved invoice	A complete invoice, timesheet included

Totals and “next due” answers are computed live from these facts, never stored, so they cannot drift out of date.

4. Sound

A single app-wide sound switch remembers on / off between launches. Every button wraps its action in a small helper that plays a sound (unless muted) and then runs the action, so screens never check the setting themselves. If playback fails, it is silently ignored — sound is never allowed to break anything.

Sound	Used for
Click	Ordinary taps: navigation, toggles, dialog buttons
“Select” chime	Moments that actually save something — adding a bank, member, note and so on — so the sound hints that something was committed
Error click	A form that failed validation

5. Look-and-Feel Components

Component	What it does
Gradient header bar	The dark navy-to-charcoal top bar with a soft green glow, an icon badge, an optional subtitle, round glassy action buttons and an automatic back arrow when there is somewhere to go back to
Footer navigation	Five tabs — Vault, Notes, Home, Log, Stash — with a raised round Home button that looks fused to the bar. Tab numbers are fixed across the app; only the labels were once renamed
Attention badge	The red / yellow circle with a “!” or a tick, either inline on a row or floating at a tile’s corner; draws nothing when all is clear

6. Input Components

Component	What it does
Inline date picker	Tap a field and a compact calendar opens right underneath it. A month / year wheel allows long jumps (such as a birth year decades ago), and a “month and day only” mode serves things that repeat every cycle
Inline time picker	The time-of-day sibling: tap the dial of an analog clock, or switch to a simple up / down digital spinner
Suggestion text box	Remembers past entries and offers up to five matches as you type — “starts with” matches first, then “contains” — and shows nothing if the box is empty or nothing matches
Rich-text engine and toolbar	Tracks which parts of a note are bold, italic, underlined or struck-through, and powers bullet lists and line breaks
Biometric helper	Asks the phone whether fingerprint / face unlock is available and runs the phone’s own prompt; any error counts as “not available” so it can never block the normal lock

7. Date-Maths Conventions

- **Month-end clamping.** Adding months to a date that doesn’t exist in the target month (31st into February) lands on the last real day of that month.
- **Day-only comparisons.** Maturity and due dates are compared by calendar day, ignoring the time.
- **Fresh, not stored.** Wherever possible, dates and totals are recomputed from raw facts every time they are needed.